Gesture-Based Assistive Communication Using TinyML and FSK for Real-Time Embedded Systems

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***Abstract*—The proposed gesture-based assistive communication system uses the concept of translating gestures into audible speech through the processing of the gestures using a real-time embedded system. The proposed system uses an MPU6050 sensor to capture the gestures. The MPU6050 sensor consists of an accelerometer and a gyroscope. The proposed system uses the ESP32 microcontroller along with the MPU6050 sensor to capture the gestures. The TinyML gesture recognition model is deployed on the ESP32 microcontroller. The TinyML model recognizes the gestures and sends the detected gestures to the MATLAB platform for processing. The detected gestures are converted into text messages and are sent to the text-to-speech interface to generate the audible speech. The proposed system uses the concept of digital communication by encoding the detected gestures into binary format and using the FSK modulation technique for the analysis and visualization of the communication system. The proposed system uses the TinyML technique along with the concept of digital communication to develop an efficient communication system.**

Keywords: TinyML, Gesture Recognition, Gesture-to-Speech System, FSK Modulation.

I. Introduction

The rapid advancement in wearable technology and embedded sensor systems has led to new opportunities for improving human-computer interaction. Among these emerging technologies, gesture-based interaction systems have gained significant popularity due to their ability to offer users a more natural way of interacting with computing systems. A prime example of this trend is the Gesture-Based Assistive Communication System, which utilizes motion sensors, TinyML, and FSK communication protocols for recognizing hand gestures and converting them into meaningful information in the form of text or speech.

The main component of this communication system is a motion sensor known as MPU6050, which consists of both an accelerometer and a gyroscope for accurately recognizing hand gestures. The sensor is embedded in a wearable device for recognizing hand gestures performed by users. The ESP32 microcontroller is used for processing sensor information, utilizing TinyML for recognizing gestures, and sending information through a Frequency Shift Keying (FSK) protocol for communication with a receiver. At the receiving end, a Python-based interface is used for visualizing gestures and converting them into speech for communication by users who suffer from speech or motor disorders.

The importance of the system is based on the fact that it is capable of providing an affordable, portable, and real-time assistive communication system that can be used entirely offline without the need for cloud-based computation. This project is based on the integration of sensor technology, machine learning, and digital communication, providing a platform that can be used to improve the life of people with disabilities.

This study is based on the exploration of the technological aspects of the system, including the sensor integration, sensor processing, gesture recognition using TinyML, and FSK-based communication. In addition, the study is based on the exploration of the potential applications of the system in the real world, including the importance of wearable devices in the real world and the significance of innovation in wearable technology.

II. Literature Review

Over the past few years, there has been substantial research in the development of wearable technology for gesture recognition, especially for assistive communication purposes. In the initial stages of the study, there was the usage of gyroscopes and accelerometers for the detection of hand movements. This was followed by the analysis of the data collected from the sensors to recognize the gestures. Some of the examples of the usage of gesture recognition include the development of data gloves and the Wii Remote. However, there have been challenges in the accuracy of the results obtained from the sensors.

The development of sensor technology has been one of the most important factors in the development of the precision of gesture recognition systems. In recent times, the MPU6050 sensor has been widely used for the purpose of gesture recognition. It is a combination of the accelerometer and the gyroscope sensor. It has been observed that the usage of sensor fusion techniques and the development of proper programming algorithms have resulted in the precision of the results obtained from the sensors.

At the same time, the developments in machine learning, especially lightweight machine learning, have enabled the shift from the offline post-processing of gestures to real-time computation. TinyML has enabled the ESP32 microcontroller to perform complex computations on the embedded system itself, eliminating the constant need for connectivity to the cloud. Various recent studies have proven the effectiveness of the integration of TinyML with wearable devices not only

in terms of real-time feedback but also in terms of the cost-effectiveness of the assistive system.

Recent studies have highlighted the significance of real-time feedback in the development of assistive systems based on gestures. Mobile applications have been designed to provide real-time feedback to the user in the form of visual and auditory feedback. Various studies have proven the effectiveness of the combination of wearable devices with interactive software in the development of assistive systems for people with disabilities. It has been proven that gestures can be translated into text or speech through the combination of wearable devices with interactive software.

Overall, these developments highlight the promise of sensor technology, machine learning, and interactive technology in delivering advanced assistive technology solutions. Leveraging previous research in motion sensing, embedded technology, and feedback, gesture recognition technology can now deliver more accurate, user-friendly, and pragmatic solutions for facilitating communication for those with impairments.

III. Methodology

A. System Architecture

The gesture-to-speech communication system, as proposed, utilizes the MPU6050 gesture sensor, ESP32 microcontroller, and MATLAB processing environment for real-time communication. The MPU6050 gesture sensor utilizes its in-built accelerometer and gyroscope to detect hand gestures based on its acceleration and angular velocity. The ESP32 microcontroller is connected to the MPU6050 gesture sensor using the I2C communication protocol and continuously reads the data from the hand gestures.

The ESP32 microcontroller is programmed using the Arduino Integrated Development Environment and is used for initial data processing, such as noise removal and normalization. A TinyML gesture recognition model is also integrated with the ESP32 microcontroller for real-time gesture recognition. Once a valid gesture is recognized by the model, the gesture information is sent to the computer system using serial communication.

The gesture information is processed in the MATLAB environment and converted to a predefined text message based on the gesture recognition. The text message is converted to a speech signal using the Windows Speech Synthesizer interface. In addition, the text message is converted to a binary format and modulated using the Frequency Shift Keying (FSK) technique. MATLAB tools, such as time domain and spectrogram plots, are used for analyzing the gesture and communication signals.

B. Block Diagram

The block diagram of the proposed gesture-to-speech communication system illustrates an organized framework that incorporates motion sensing hardware, embedded processing, and MATLAB-based signal analysis, enabling hand gestures to be translated into sound and communication signals. This proposed gesture-to-speech communication system incorporates

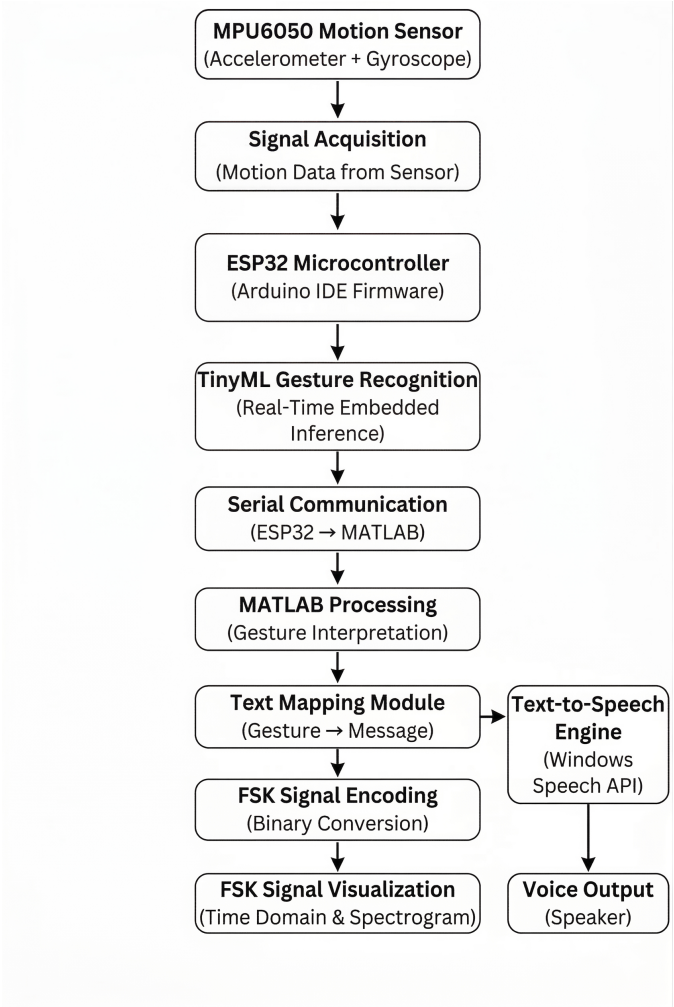

Fig. 1. System Architecture of Gesture-to-Speech Communication

an MPU6050 motion sensing hardware, an ESP32 microcontroller, and MATLAB-based processing.

At the sensing level, an MPU6050 motion sensing hardware is utilized, and it captures hand gesture information. This MPU6050 motion sensing hardware incorporates an accelerometer and a gyroscope, and these two sensors can effectively detect linear and angular movements in the hand. This information provides detailed information about hand gestures, including orientation and dynamic motion characteristics. This sensor continuously measures raw data that corresponds to hand gestures made by users.

The ESP32 microcontroller is an embedded processing hardware that functions as the core processing element in the proposed gesture-to-speech communication system. It communicates directly with the MPU6050 motion sensing hardware via an I2C communication protocol and captures motion data in real time. This microcontroller is programmed using an Arduino IDE.Within the firmware, the raw sensor readings are filtered to eliminate any noise and stabilize the signal measurements. Basic preprocessing and calibration are performed on the raw sensor readings to ensure accurate representation of

the motion data.

Once the preprocessing is complete, the ESP32 sends the captured gesture data to a computer system. The ESP32 uses a serial communication method to send the sensor readings to the MATLAB system. Within the MATLAB system, the captured gesture signals are processed. The waveform of the motion is analyzed to observe the gestures. An energy detection method is employed to identify the occurrence of a valid gesture. Within the MATLAB system, the energy of the signal is calculated and compared with a predefined threshold value to differentiate gestures from idle states.

Upon the occurrence of a valid gesture, the system classifies the gestures by mapping the identified motion pattern to a predefined text message. Every gesture is mapped to a predefined word or command. This process converts the gestures into a digital text format that can be utilized for communication. The recognized text message is then processed in a speech synthesizer module. MATLAB interacts with the Windows Speech Synthesizer using the .NET framework and incorporates a text-to-speech (TTS) engine. The text message is converted into speech and played through the system speaker.

In addition to speech output, the system also demonstrates the generation of a communication signal using the text message in binary form using ASCII code. The binary message is modulated using the frequency shift keying (FSK) technique. In the FSK technique, two different carrier frequencies represent binary '1' and binary '0.' MATLAB generates the FSK waveform corresponding to the binary message and displays the modulated signal in the time domain.

The generated signal can be further analyzed using a spectrogram visualization tool. The tool shows the frequency transitions during the modulation process. It can be noted that the frequency transitions during the modulation process are valid, proving the implementation of the FSK technique.

The above description illustrates a block diagram of a system that incorporates motion sensing hardware, embedded systems using Arduino-based firmware, and MATLAB-based signal processing. It demonstrates the feasibility of a gesture-based speech communication system and the potential of using embedded systems with high-end signal processing platforms.

The methodology adopted for developing the gesture-to-speech communication system is as follows:

Firstly, during the signal generation phase, a simulated waveform is generated for the gesture motion within a MATLAB environment. Since the system is intended to mimic a signal generated by accelerometers for motion detection, similar to those generated by MPU6050 devices, a sinusoidal waveform is adopted along with some level of random noise to mimic hand movement patterns. The above-mentioned simulated waveform is intended to mimic a realistic signal that could be generated by motion sensors like MPU6050 devices.

Secondly, during the signal processing and gesture detection phase, the system is intended to analyze the motion signal generated earlier to ascertain whether a gesture has been performed or not. The MATLAB environment calculates the total amount of energy within the signal by summing up squared values within the signal. If the amount of energy is found to be greater than a predefined threshold value, it is ascertained that a gesture has been performed; otherwise, it is ascertained that an idle signal is being generated.

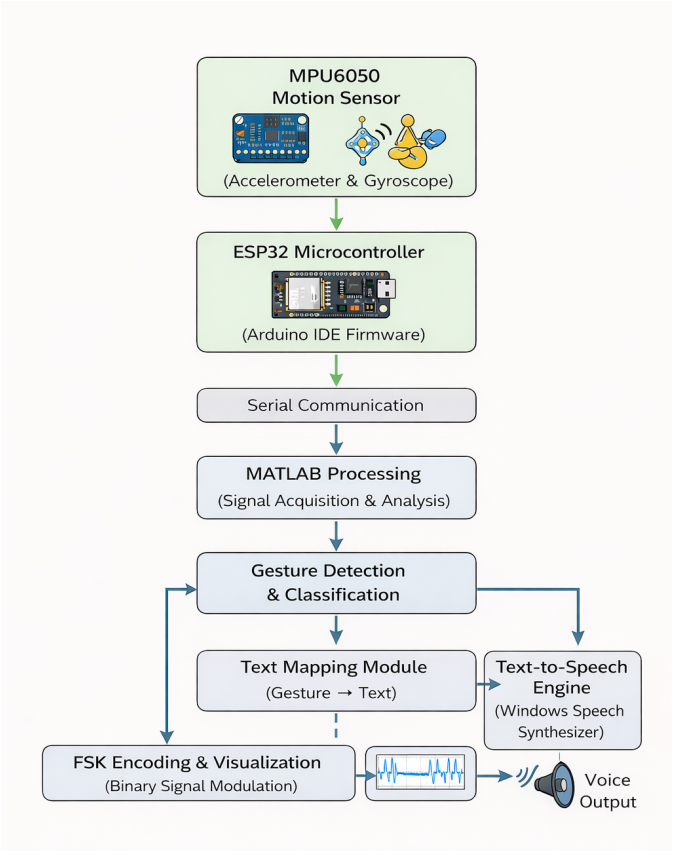

Fig. 2. Block Diagram Of Gesture-To-Speech Communication

Thirdly, the classification of the gestures and the text mapping are done once a valid gesture is detected. The detected gesture is mapped to a predefined word or message stored in the system. This process converts the detected gestures into text that can be processed for communication purposes.

Fourthly, the speech synthesis is done by integrating the Windows Speech Synthesizer tool with MATLAB. The text mapped from the detected gestures is sent to the text-to-speech tool. The text is converted into voice using the text-to-speech tool. This enables the system to generate voice corresponding to the detected gestures. This helps the system to communicate through gestures.

Lastly, the system uses the concept of digital communication by converting the text generated from the detected gestures into binary format. The text is modulated using Frequency Shift Keying (FSK) to generate a modulated waveform. In FSK modulation, the binary values are represented by two different frequencies. MATLAB is used to generate the modulated waveform. The waveform is analyzed by plotting the time-domain plots and the spectrogram.

Testing and visualization are conducted to ensure proper

operation of each stage. MATLAB plots are employed to analyze gesture signals, test the performance of the proposed energy detection method, and check the frequency transition of generated FSK signals. By passing through these stages, it is evident that the proposed system is able to convert gesture signals into speech as well as digital communication signals.

IV. Results And Discussion

A. Hardware and Software Integration

In the proposed system, gesture sensing, embedded processing, and speech generation are integrated to enable real-time gesture-to-speech communication. The system utilizes an MPU6050 motion sensor connected to an ESP32 microcontroller to capture hand motion data using accelerometer and gyroscope measurements. The ESP32 is programmed using the Arduino IDE and performs initial signal preprocessing, including noise filtering and normalization of the sensor data. A TinyML-based gesture recognition model is deployed on the ESP32 to classify hand gestures in real time directly on the embedded device. Once a gesture is detected, the recognized gesture information is transmitted to the MATLAB environment through serial communication. MATLAB processes the received gesture data and converts the recognized gesture into a predefined text message. The text message is then converted into speech using the Windows Speech Synthesizer interface. In addition, the message is encoded into binary data and modulated using the Frequency Shift Keying (FSK) technique for signal analysis.

B. Gesture Recognition and System Performance

Gesture recognition in the proposed system is performed using a TinyML-based model deployed on the ESP32 microcontroller. The model analyzes motion data obtained from the MPU6050 sensor and classifies the gestures into predefined categories. Experimental observations show that the system can successfully distinguish between active gestures and idle states using the trained TinyML model. The integration of embedded machine learning enables real-time gesture recognition with low computational overhead. Once the gesture is identified, the corresponding message is generated and transmitted for further processing. The generated binary message is modulated using the FSK technique, where two different frequencies represent binary data. The resulting FSK signal clearly demonstrates frequency switching corresponding to the transmitted binary sequence, confirming the correct implementation of the modulation technique.

C. Voice Output and System Interaction

After the gesture is successfully recognized, the corresponding text message is converted into speech using MATLAB's integration with the Windows Speech Synthesizer. The generated speech is played through the system speaker, providing real-time voice feedback to the user. This demonstrates the capability of the system to translate hand gestures into understandable spoken communication. The integration of gesture sensing, TinyML-based recognition, and speech generation highlights the feasibility of developing assistive communication systems for individuals with speech or hearing impairments. The results indicate that the proposed system can effectively detect gestures and generate voice output, enabling intuitive and accessible human–machine interaction.

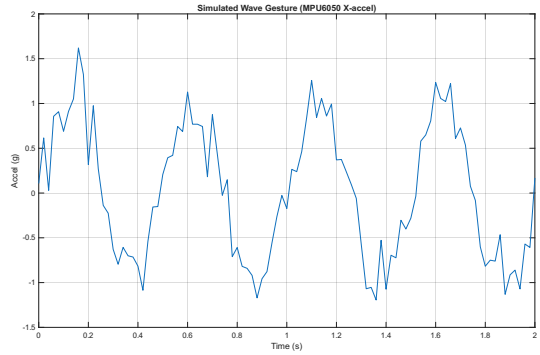

Fig. 3. Simulated Gesture Signal From MPU6050 Accelerometer

V. Conclusion

The proposed gesture-to-speech communication system incorporates hardware and software components to recognize hand gestures and convert them into communicable speech in real time. The proposed communication system incorporates an MPU6050 motion sensor to collect hand gesture data using accelerometer and gyroscope readings. The ESP32 microcontroller processes hand gesture data to recognize gestures using a TinyML-based gesture recognition model. The recognized hand gestures are then transmitted to a MATLAB processing environment to convert them into predefined text messages and speech using a text-to-speech engine.

The proposed communication system is not only able to convert hand gestures into communicable speech, but it is also able to demonstrate digital communication capabilities by encoding the generated message using Frequency Shift Keying (FSK) modulation techniques. The experimental results demonstrate that the proposed communication system is able to accurately recognize hand gestures and convert them into communicable speech with minimal delays. The integration of TinyML-based gesture recognition capabilities into the proposed communication system is efficient and accurate.

The proposed communication system demonstrates its capabilities to incorporate efficient communication systems using embedded systems and signal processing techniques to enable efficient communication systems that can greatly improve communication accessibility for people with speech and hearing difficulties.

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